Vocabulary by topic

389 existing reference entries, organized to accompany the ten-page reading guide.

Use this as a reference when a word interrupts your reading. The core guide explains essential terms in context; this companion retains the fuller collection. Entries are grouped by broad topic and source association, rather than alphabetically. Some terms connect several topics.

Reference status

Definitions and source labels are inherited from an existing vocabulary collection and have not all been independently reverified. They may include source-specific proposals, contested interpretations, or historical claims. A link is a bookmark for checking the original source, not a guarantee that every statement in the note has been validated. Podcast references do not supply verified timestamps.

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01 / Three questions, one reading journey

5 inherited entries. Corresponds to page 1 of the reading guide. Original note order is retained within this topic.

position paper

A paper that argues a vision and does not report tested results; LeCun calls 06.3 this, so its claims are proposals to test

Inherited source reference: 06.3

hierarchical planning is unsolved

LeCun calls it a largely unsolved problem; most solutions need a predefined action vocabulary, whereas he wants intermediate action representations to be learned

Inherited source reference: 06.3

open problems in 06.3

Unproven: whether H-JEPA can be built and trained from video; how best to regularize the latent; how the actor infers actions and latents; how short-term memory represents beliefs; and the configurator, which he calls the most mysterious module; he expects many years of work

Inherited source reference: 06.3

PEVA limitations

Planning covers only simulated arm actions, no long-horizon or full-trajectory optimization, no closed-loop control, no explicit semantic goals, and image similarity is only a proxy objective; the authors call it an initial step

Inherited source reference: 06.10

bitter lesson

Richard Sutton's argument that general methods that scale with computation have beaten approaches built on human domain knowledge, so leveraging computation is what matters in the long run; 06.7 argues for a more nuanced view

Inherited source reference: 06.7

02 / From observations to a world model

60 inherited entries. Corresponds to page 2 of the reading guide. Original note order is retained within this topic.

autonomous machine intelligence

LeCun's goal: machines that learn mostly by observation, reason and plan, and are driven by built-in objectives rather than hard-wired programs or external rewards

Inherited source reference: 06.3

world model

A learned internal model of how the world works; it fills in missing information and predicts plausible future world states, including those that follow a proposed sequence of actions

Inherited source reference: 06.3

common sense

In 06.3, a collection of world models that tell an agent what is likely, plausible or impossible; the basis for learning new skills in few trials and avoiding dangerous mistakes

Inherited source reference: 06.3

configurator

Executive-control module: given a task, it pre-configures perception, world model, cost and actor by modulating their parameters and attention

Inherited source reference: 06.3

perception module

Estimates the current state of the world from sensor signals, possibly at several levels of abstraction, keeping what is relevant to the task

Inherited source reference: 06.3

actor module

Proposes sequences of actions to the world model, optimizes them against the cost, and outputs the first action to the effectors

Inherited source reference: 06.3

cost module

Computes one scalar, the energy, measuring the agent's discomfort; it is the sum of the intrinsic cost and the trainable critic; the agent aims to stay in low-energy states

Inherited source reference: 06.3

intrinsic cost

The hard-wired, non-trainable part of the cost module: instantaneous pain, pleasure, hunger and similar drives; its design shapes the agent's behavior

Inherited source reference: 06.3

critic (trainable critic)

Trainable part of the cost module that predicts future intrinsic cost, learning from past states and costs retrieved from memory

Inherited source reference: 06.3

energy

In 06.3, a scalar measure of the agent's discomfort (low is good); the agent acts to minimize its average over time

Inherited source reference: 06.3

short-term memory

Stores past, current and predicted world states with their intrinsic costs; the world model reads and updates it; compared to the vertebrate hippocampus

Inherited source reference: 06.3

key-value memory network

A memory design the paper says short-term memory may resemble: stored values are retrieved by matching a query to keys

Inherited source reference: 06.3

effectors

The outputs that act on the world (motors, actuators) that receive the actor's chosen action

Inherited source reference: 06.3

latent variable

A hidden variable that stands for what is unknown or uncertain, letting a world model output several plausible futures instead of one

Inherited source reference: 06.3

levels of abstraction

Representing the world coarsely or finely; higher levels ignore hard-to-predict detail and allow longer-term prediction

Inherited source reference: 06.3

uncertainty (multiple plausible predictions)

The world is not fully predictable, so a world model must represent many possible outcomes, not one

Inherited source reference: 06.3

intuitive physics

Basic expectations (gravity, inertia, object permanence) infants learn mostly by observation, with higher concepts built on lower ones

Inherited source reference: 06.3

configurable world-model engine

The hypothesis that people and animals have one dynamically configurable world model, so knowledge is shared across tasks and enables reasoning by analogy

Inherited source reference: 06.3

cognitive architecture

A design for how perception, memory, prediction, cost and action modules fit together; 06.3 proposes one in which all modules are differentiable

Inherited source reference: 06.3

aleatoric uncertainty

Uncertainty because the world is itself random, so one state and action can lead to several outcomes

Inherited source reference: 06.3

epistemic uncertainty

Uncertainty because the agent's state representation or training data is incomplete, so predictions are imperfect

Inherited source reference: 06.3

factor graph

A probabilistic graphical model; 06.3 says the proposed architecture is one, with the cost modules acting as log factors

Inherited source reference: 06.3

behavioral drives

Submodules of the cost (pain, hunger, curiosity, social drives, standing up to walk) whose weights the configurator adjusts so the agent focuses on different subgoals at different times

Inherited source reference: 06.3

associative memory

A memory retrieved by content, not address; here the intrinsic cost stores (time, state, energy) triplets and the critic reads them back

Inherited source reference: 06.3

three world-model challenges

(1) diverse training data, since what is seen depends on actions taken; (2) representing multiple plausible outcomes; (3) predicting at several time scales and abstraction levels

Inherited source reference: 06.3

energy-based model (EBM)

A scalar function F(x,y) that is low when x and y are compatible and high when not; it can represent cases where many y fit one x, without having to generate y

Inherited source reference: 06.3

multi-modal dependency

A situation where several different outputs are all compatible with one input, such as many plausible continuations of a video clip

Inherited source reference: 06.3

energy landscape

The surface of an EBM's energy values: low near data points, high away from them; contour lines mark level sets

Inherited source reference: 06.3

emergent concepts from video (depth, objects)

LeCun's hypothesis: predicting video in representation space would make edges, depth maps, occlusion and object permanence emerge, building a hierarchy of concepts

Inherited source reference: 06.3

latent-variable energy-based model (LVEBM)

An energy model E(x,y,z) with a hidden variable z that parameterizes the possible relationships between x and the y that fit it; inference picks the z that minimizes energy

Inherited source reference: 06.3

latent variable inference

Finding the hidden z that best explains a pair (x,y) by minimizing energy (e.g. the camera motion between two views, or whether a car turned left or right); eliminating z leaves F(x,y)

Inherited source reference: 06.3

collapse-prone architectures

Per 06.3: plain predictors cannot collapse; generative latent-variable models collapse if z has too much capacity; auto-encoders collapse if the code is large enough to learn the identity; joint embeddings collapse if encoders output constants

Inherited source reference: 06.3

auto-encoder

A network that compresses an input to a code and reconstructs it; it can collapse by learning the identity function when the code has too much capacity

Inherited source reference: 06.3

joint embedding architecture (JEA)

Two encoders map x and y to representations that are pushed to agree; can collapse if the encoders ignore inputs and emit identical constants

Inherited source reference: 06.3

curse of dimensionality

The way requirements blow up as data dimension grows; LeCun's main reason to argue against contrastive methods

Inherited source reference: 06.3

Siamese network

Twin networks with shared weights trained on pairs (a distorted version versus another sample); the architecture behind many contrastive methods

Inherited source reference: 06.3

JEPA (Joint Embedding Predictive Architecture)

Two encoders turn x and y into sx and sy; a predictor (optionally using a latent z) predicts sy from sx; energy is the prediction error in representation space; non-generative, so it need not predict every detail of y

Inherited source reference: 06.3

H-JEPA (hierarchical JEPA)

JEPAs stacked so higher levels take lower-level representations and make longer-term, more abstract predictions; the basis for hierarchical planning under uncertainty

Inherited source reference: 06.3

encoder invariance

An encoder that maps many different inputs to the same representation, so they all get the same energy; one way a JEPA handles many compatible outcomes and drops irrelevant detail

Inherited source reference: 06.3

representation-space prediction

Predicting an abstract encoding of the future instead of its raw pixels; ignores irrelevant detail like trees by the road or sidewalk texture

Inherited source reference: 06.3

variational auto-encoder (VAE)

A generative auto-encoder with a noisy latent whose distribution is regularized; an example of limiting latent information

Inherited source reference: 06.3

energy not probability

An EBM is defined without probabilities; the energy is the fundamental object, though it can be turned into a probability via a Gibbs distribution

Inherited source reference: 06.3

expander

A small trainable network that maps representations to higher-dimensional embeddings before VICReg's variance and covariance losses are applied

Inherited source reference: 06.3

inductive bias

The built-in preference set by encoder and predictor architecture about what information counts as predictable; extra prediction heads for task-relevant variables can steer it

Inherited source reference: 06.3

case against generative models

LeCun argues generative latent-variable models cannot drop irrelevant detail except by pushing it into a latent, because they do not form abstract invariant representations of y; predicting every pixel (leaves, ripples, carpet texture) is impractical

Inherited source reference: 06.3

multi-scale prediction

Low-level representations carry detail for short-term prediction; high-level ones are coarser and enable long-term prediction; temporal pooling between levels coarse-grains the representation

Inherited source reference: 06.3

types of uncertainty (aleatoric 1-3, epistemic 1-4)

06.3 lists: intrinsic randomness; deterministic but chaotic; deterministic but partly observable; sensors give partial information; representation lacks needed information; model limited by representational power (bounded rationality); model trained on limited data

Inherited source reference: 06.3

bounded rationality

An agent's model being inaccurate because of limits on its representational power or computation

Inherited source reference: 06.3

gating and dynamic routing

Mechanisms where latent variables modulate which connections are used between frames or modules; suggested for the world model's low-level video prediction

Inherited source reference: 06.3

world-state memory (key-value, differentiable)

A memory where a query is matched to keys, normalized, and used to read or update stored values; each entry can be the state of one entity (a bottle, the kitchen); all steps are differentiable

Inherited source reference: 06.3

actor's three roles

Infer optimal action sequences; produce configurations of latent variables for the unknown part of the world; train policy networks for Mode-1; in adversarial settings the latents are chosen to maximize cost

Inherited source reference: 06.3

configurator (design)

Main controller that modulates other modules' parameters and connection graphs (routing signals, activating sub-networks, focusing attention); with transformers, its outputs can act as extra input tokens; needed for hardware reuse and knowledge sharing

Inherited source reference: 06.3

transformer permutation equivariance

Transformer blocks give the same result however objects are assigned to input tokens, which suits reasoning about interacting objects; extra input tokens can reconfigure the connection graph

Inherited source reference: 06.3

causal attention

Constraining attention to the past so a transformer predicts the next state, action and cost without seeing the future

Inherited source reference: 06.3

Director (Hafner 2022)

A related hierarchical world-model and planning system trained end-to-end through reinforcement learning, cited as prior work on hierarchical planning

Inherited source reference: 06.3

brain analogies

06.3 maps modules to brain areas: perception to sensory cortex, world model and critic to prefrontal cortex, intrinsic cost to amygdala, short-term memory to hippocampus; explicitly speculative

Inherited source reference: 06.3

EBM (broad sense)

In 06.3 EBM means any model treating the energy function as fundamental and shaping its landscape by learning, broader than the probabilistic definition; the Boltzmann machine is an early contrastively trained example

Inherited source reference: 06.3

physical grounding

The audio's term for models tied to real physical cause and effect, which text-only language models are said to lack; an idea from the world-model argument, not a measured property

Inherited source reference: audio

spatial persistence and physical causality

The audio's examples of what an internal causal model must capture: objects stay where they are and actions have physical consequences; related to intuitive physics and object permanence in 06.3

Inherited source reference: audio; cf. 06.3

composite architecture

The program summary's and audio's picture of future AI: language fluency plus perception, persistent memory, a world model and possibly symbolic reasoning, as modules, not one giant network; a hypothesis, not an established design

Inherited source reference: audio; GLAF summary

03 / Learning useful representations

32 inherited entries. Corresponds to page 3 of the reading guide. Original note order is retained within this topic.

self-supervised learning (SSL)

Learning without labels by predicting parts of the data from other parts; 06.3 calls training a world model a prototypical case of pattern completion

Inherited source reference: 06.3

intrinsic motivation

Behavior driven by built-in objectives (drives such as curiosity or avoiding pain) rather than external reward; set by the intrinsic cost module

Inherited source reference: 06.3

differentiable module

A module that can pass back gradient estimates of the cost, so the whole system can be trained and can plan by gradient methods

Inherited source reference: 06.3

policy gradient methods

Classical reinforcement-learning approach: estimate how the cost changes by trying slightly perturbed actions in the world; the slow, risky route a world model is meant to avoid

Inherited source reference: 06.3

amortized inference

Train a fast policy network to approximate the result of slow Mode-2 planning, so good actions come from one pass; the agent "compiles" planned skills into reactive behavior

Inherited source reference: 06.3

skill compilation (Mode-2 to Mode-1)

Using slow deliberate planning to discover an action, then training the reactive policy on it by minimizing a divergence from the planned action; the agent can focus on only one complex task at a time

Inherited source reference: 06.3

inverse reinforcement learning

Infer the objective an expert appears to be optimizing from watching its behavior, producing a critic for Mode-2 behavior

Inherited source reference: 06.3

imitation learning

Learning behavior from an expert; 06.3 lists it as one of four ways to specify an agent's behavior (program it, define an objective, supervise a policy, or imitate)

Inherited source reference: 06.3

objective-driven behavior

Specifying an objective rather than a behavior; 06.3 says this is simpler to engineer and more robust, since the agent can adapt to unexpected conditions

Inherited source reference: 06.3

A2C

An actor-critic reinforcement-learning method; 06.3 says its critic training resembles how the critic here is trained

Inherited source reference: 06.3

one-shot learning

Learning from a single example; 06.3 says a key-value memory with good slot allocation can do this and still interpolate between keys

Inherited source reference: 06.3

pattern completion

Given part of an input plus a proposed rest, judge whether the completion is plausible; the basic idea of self-supervised learning in 06.3

Inherited source reference: 06.3

joint embedding learning principle

Learn representations sx and sy of two inputs so both are maximally informative about their inputs and sy is easy to predict from sx, trading predictability against information

Inherited source reference: 06.3

energy collapse

Training failure where the energy surface goes flat and every y gets the same low energy, so the model says nothing; the core failure that non-contrastive methods must prevent

Inherited source reference: 06.3

contrastive methods

Train by pushing energy down on real pairs and up on made-up "contrastive" pairs; 06.3 argues the number of contrastive samples can grow exponentially with dimension, so it scales badly

Inherited source reference: 06.3

regularized (non-contrastive) methods

Push energy down on real data while a regularizer shrinks the volume of low-energy space, "shrink-wrapping" the data; LeCun says these are more promising long-term and can be combined with contrastive ones

Inherited source reference: 06.3

InfoNCE / hinge loss

Examples of contrastive loss functions: hinge loss with a distance-dependent margin; InfoNCE compares one true pair against K contrastive samples

Inherited source reference: 06.3

contrastive method examples

06.3 lists Siamese nets, DrLIM, PIRL, MoCo, SimCLR, GANs, denoising and masked auto-encoders, and contrastive divergence as contrastive or near-contrastive approaches

Inherited source reference: 06.3

VICReg / Barlow Twins

Non-contrastive criteria for training JEPAs that keep representations informative and prevent collapse without contrastive samples

Inherited source reference: 06.3

four JEPA training criteria

Maximize information in sx about x; maximize information in sy about y; make sy predictable from sx; minimize information in the latent z; the first, second and fourth prevent collapse

Inherited source reference: 06.3

informational collapse

Collapse where representations carry little information (constant codes) or where the latent can simply copy the answer; prevented by maximizing representation information and limiting latent capacity

Inherited source reference: 06.3

limiting latent information

Ways to stop z from doing all the work: make it discrete (VQ-VAE), low-dimensional, sparse (L1 norm) or noisy (VAE-style entropy)

Inherited source reference: 06.3

sparse coding / L1 regularization

Penalize the absolute size of a vector's entries so most are zero; restricts the low-energy region to low-dimensional manifolds

Inherited source reference: 06.3

VICReg (variance-invariance-covariance regularization)

A non-contrastive JEPA training method: an invariance term (prediction error) plus a variance term (keep each embedding component's spread above a threshold) and a covariance term (push correlations between components to zero), computed over a batch

Inherited source reference: 06.3

dimension-contrastive vs sample-contrastive

LeCun's contrast: VICReg makes the components of representations differ over a batch, while classic contrastive methods make different input samples differ and need many negative samples

Inherited source reference: 06.3

prediction heads

Extra output layers trained to predict task-relevant, easily derived variables, used to bias a JEPA toward useful representations

Inherited source reference: 06.3

reparameterization trick

Moving a prediction's randomness into a latent variable so it can be optimized or sampled; 06.3 does not need it because latent parameterization is fundamental to its design

Inherited source reference: 06.3

latent regularizer (R(z))

A penalty on the latent variable that limits its information; at planning time latents are sampled from the distribution it defines (via a Gibbs distribution), and the regularizer can be predicted from past states

Inherited source reference: 06.3

five modes of information gathering

Passive observation; active foveation (steering attention); passive agency (watching others act); active egomotion (moving sensors); active agency (acting, which raises the exploration-exploitation dilemma)

Inherited source reference: 06.3

exploration-exploitation dilemma

The trade-off between trying new things to learn and using what already works; sharpest when the agent's own actions change what it observes

Inherited source reference: 06.3

curiosity-driven exploration

Intrinsic motivation modules that drive attention and exploration toward parts of the state space where the world model is inexact or uncertain

Inherited source reference: 06.3

latent amortized inference module

A network that learns distributions of latent variables so good, plausible latents can be sampled quickly instead of searched

Inherited source reference: 06.3

04 / How a model becomes a plan

42 inherited entries. Corresponds to page 4 of the reading guide. Original note order is retained within this topic.

model-predictive control (MPC)

Classical control method (since the 1950s): predict the future with a forward model, pick the action sequence with lowest cost, execute the first action, repeat

Inherited source reference: 06.3

forward model

A model that predicts the next state of the world from the current state and a considered action

Inherited source reference: 06.3

Mode-1 (reactive behavior)

A perception-action episode where a policy maps the state straight to an action with no world-model prediction; likened to Kahneman's System 1

Inherited source reference: 06.3

Mode-2 (reasoning and planning)

A perception-action episode that plans through the world model and cost, like MPC; likened to System 2; reasoning here means energy minimization

Inherited source reference: 06.3

policy module

The part of the actor that directly produces an action from the estimated world state without deliberate planning

Inherited source reference: 06.3

action optimizer

The part of the actor that finds the action sequence minimizing estimated cost, using gradient methods or dynamic programming for discrete actions

Inherited source reference: 06.3

dynamic programming

A method to find an optimal action sequence by solving smaller sub-problems; 06.3 uses it when the action space is discrete

Inherited source reference: 06.3

reasoning as energy minimization

LeCun's broad sense of reasoning: constraint satisfaction, finding states or actions that minimize energy

Inherited source reference: 06.3

non-differentiable world

The real world cannot be back-propagated through, so cost gradients cannot flow from outcomes back to actions; they must be estimated by trying perturbed actions, which is slow and potentially dangerous

Inherited source reference: 06.3

Mode-2 loop (perceive, propose, simulate, evaluate, plan, act, store)

Perceive the state; the actor proposes an action sequence; the world model simulates the resulting states; the cost scores them; gradients refine the actions; the first action is executed; states and costs are stored

Inherited source reference: 06.3

receding-horizon planning

Plan a sequence of T steps ahead, execute only the first action, then replan from the new state; 06.3's loop is classical MPC except the world model and cost are learned

Inherited source reference: 06.3

gradient-free planning (tree search, simulated annealing)

Alternatives to gradient-based optimization of actions, needed when the action-cost mapping has discontinuities or actions are discrete: dynamic programming, combinatorial optimization, simulated annealing, tree search with pruning

Inherited source reference: 06.3

reasoning by simulation and analogy

LeCun's claim that planning by simulating with a world model and optimizing the energy goes beyond classical logical or probabilistic reasoning

Inherited source reference: 06.3

hierarchical planning

Humans plan at an abstract level and decompose high-level goals into subgoals, down to muscle control; a world model needs to represent plans at several time scales and abstraction levels

Inherited source reference: 06.3

hierarchical planning (top-down)

A high-level cost sets abstract actions that act as conditions or subgoals for the lower level, which then finds low-level actions to satisfy them; optimizing all levels jointly is better than the greedy top-down pass

Inherited source reference: 06.3

abstract action

In hierarchical planning, a target or condition for the level below to satisfy, not a real motor command; old idea in control theory

Inherited source reference: 06.3

proportional servomechanism

A classical controller given a target state whose control is proportional to the negative gradient of the squared distance to that target

Inherited source reference: 06.3

trajectory explosion

With k possible discrete latent values and t time steps the number of predicted trajectories grows as k^t; directed search and pruning such as Monte-Carlo tree search are needed

Inherited source reference: 06.3

Monte-Carlo tree search (MCTS) / beam search

Search strategies that explore only promising branches; 06.3 suggests them for discrete actions or latents; beam search and MCTS count as approximate dynamic programming

Inherited source reference: 06.3

risk-aware planning

Choosing actions that minimize the average predicted cost, or the average plus its variance so as to minimize risk or uncertainty about the cost

Inherited source reference: 06.3

gradient-free search (belief propagation, SAT, MCTS)

Options if gradient-based planning fails on discrete or non-smooth problems

Inherited source reference: 06.3

GRASP (Gradient RelAxed Stochastic Planner)

A planner for learned world models: lifts the trajectory into virtual states so optimization runs in parallel across time, adds Gaussian noise to the state iterates for exploration, and reshapes gradients to avoid brittle state-input gradients; still optimizes states, not "abandoning" them (the audio overstates this)

Inherited source reference: 06.9

world model (working definition in 06.9)

A learned model that, given current state and a sequence of future actions, predicts what happens next, approximating the environment's true conditional distribution; the post notes the term is overloaded (explicit dynamics model versus implicit internal state in a generative model, e.g. an LLM's board representation when playing chess)

Inherited source reference: 06.9

differentiable simulator

Because a world model is differentiable you can roll it forward under hypothetical action sequences and backpropagate through the predictions

Inherited source reference: 06.9

terminal-error planning (rollout)

The simplest planner: roll the model forward from the start state under an action sequence and minimize the error between the final state and the goal

Inherited source reference: 06.9

backpropagation through time (BPTT)

Differentiating through a model applied to itself repeatedly; over long horizons it leads to exploding or vanishing gradients

Inherited source reference: 06.9

ill-conditioned long-horizon graph

The Jacobian's conditioning grows exponentially with horizon T, so gradients to early actions explode or vanish; the first of three long-horizon pain points

Inherited source reference: 06.9

non-greedy landscape (local minima)

Long tasks need non-greedy moves (going around a wall, backing up) and the search space grows with horizon, so the loss surface is rough with many traps; the second pain point

Inherited source reference: 06.9

collocation (lifted-state planning)

Treat the dynamics as a soft penalty and optimize actions and states together; shares the same global minimizers as rollout planning, evaluates all T steps in parallel, and replaces a deep product of Jacobians with a sum

Inherited source reference: 06.9

state-input gradient (D_s F)

The gradient of a world model's output with respect to its state input; 06.9 calls it untrustworthy, adversarial and brittle for deep-learning models

Inherited source reference: 06.9

action gradient (D_a F)

The gradient with respect to action input; reasonable to optimize because the action space is low-dimensional and densely trained

Inherited source reference: 06.9

adversarial robustness

Tiny input changes along a model's gradient can flip its output (adversarial examples); appears in vision, LLMs and RL; there is a known trade-off between robustness and performance

Inherited source reference: 06.9

data manifold (tangential vs normal directions)

Training data lies near a low-dimensional manifold; training controls behavior along it but not in orthogonal (normal) directions, where the model can be very sharp, so states can be nudged to fool it

Inherited source reference: 06.9

Lipschitz constant

A bound on how fast a function's output can change with its input; reasonable along the data manifold, very high in normal directions

Inherited source reference: 06.9

stop-gradient dynamics loss

GRASP's loss that blocks gradients into the world model's state input; alone it fails (trivial minima), so a dense goal term is added

Inherited source reference: 06.9

dense goal term

A goal-distance penalty applied across the trajectory to restore the goal signal to earlier states; balanced against the dynamics loss so it does not become purely greedy

Inherited source reference: 06.9

state noising

Gaussian noise added to virtual state updates so the planner can hop between basins; actions stay gradient-guided; not truly Langevin dynamics because actions are not noised

Inherited source reference: 06.9

periodic sync

Every K_sync iterations GRASP rolls out from the real start state and takes a few gradient steps on the serial loss to keep states grounded; a serial planner such as CEM could replace it

Inherited source reference: 06.9

Push-T

A robot pushing-task benchmark where GRASP reports higher success and faster median time at longer horizons; the post gives no numbers in the readable text

Inherited source reference: 06.9

Markovian world model

A model whose next-state prediction depends only on a recent window of states and the current action; assumed for simplicity

Inherited source reference: 06.9

diffusion-based world models

Named as an extension: deeper diffusion timesteps can be viewed as smoothed versions of the world model

Inherited source reference: 06.9

compounding error (audio framing)

The audio's explanation that tiny per-step prediction errors multiply over long horizons; the GRASP source instead blames exploding or vanishing gradients, rough local minima and adversarial state gradients

Inherited source reference: audio; cf. 06.9

05 / Dreamer: learning through imagined experience

59 inherited entries. Corresponds to page 5 of the reading guide. Original note order is retained within this topic.

DreamerV3

A reinforcement-learning algorithm of three networks trained together from replayed experience: a world model that predicts outcomes, a critic that judges their value, and an actor that picks actions; one fixed configuration reportedly beats specialized methods on 150+ tasks

Inherited source reference: 06.5

reinforcement learning (RL)

Learning to act by interaction and reward; 06.5 notes it is also used to improve language models beyond their pretraining data

Inherited source reference: 06.5

PPO (proximal policy optimization)

A standard, widely applicable RL algorithm used as the general-purpose baseline Dreamer is compared against

Inherited source reference: 06.5

hyperparameters

Settings of a learning algorithm tuned by hand per task; needing to retune them is the brittleness 06.5 says holds RL back; Dreamer uses fixed ones

Inherited source reference: 06.5

sparse rewards

Reward signals that arrive rarely (like collecting a diamond in Minecraft), making exploration and long-horizon credit hard

Inherited source reference: 06.5

Minecraft diamond challenge

Benchmark task of collecting diamonds from scratch, hard due to sparse rewards, exploration difficulty, long horizons and procedural worlds; earlier methods used human data or curricula

Inherited source reference: 06.5

learning from imagination

Dreamer's actor and critic learn behavior purely from abstract trajectories of representations predicted by the world model; while acting in the environment, actions are sampled from the actor with no lookahead planning

Inherited source reference: 06.5

recurrent state-space model (RSSM)

Dreamer's world model: an encoder maps inputs to stochastic representations z; a sequence model with a recurrent state h predicts them from past actions; separate heads predict reward and episode continuation and reconstruct inputs

Inherited source reference: 06.5

model state

The combination of recurrent state h and representation z; the actor and critic operate on it, benefiting from the Markovian representations the world model learns

Inherited source reference: 06.5

autoencoding

Compressing inputs into a compact code and reconstructing them; used so the world model's representations stay informative

Inherited source reference: 06.5

straight-through gradients

A trick that lets gradients pass through a sampling step of discrete representations so the encoder can be trained end to end

Inherited source reference: 06.5

world-model losses

Three terms minimized end to end: prediction loss (reconstruct inputs, predict reward and continuation), dynamics loss (sequence model predicts the next representation) and representation loss (make representations predictable), weighted 1, 1 and 0.1

Inherited source reference: 06.5

KL divergence

A measure of how different two probability distributions are; used here to align predicted and actual representations

Inherited source reference: 06.5

stop-gradient

An operator that blocks gradient flow through one side of a loss; the dynamics and representation losses differ only in where it sits and their scale

Inherited source reference: 06.5

free bits

Clip the KL losses below 1 nat (about 1.44 bits) so they switch off once well minimized, avoiding a degenerate solution where dynamics are trivial to predict but carry little information

Inherited source reference: 06.5

unimix

Make categorical outputs a mix of 1% uniform and 99% network output so they can never become deterministic, preventing spikes in the KL losses

Inherited source reference: 06.5

symlog and symexp

symlog(x)=sign(x)ln(|x|+1) squashes large positive and negative values while staying near the identity around zero; symexp is its inverse; used for observations and prediction targets so scale does not wreck learning

Inherited source reference: 06.5

symexp twohot loss

For rewards and returns the network outputs a softmax over exponentially spaced bins and is trained with cross entropy on a twohot target, so gradient size no longer depends on target size

Inherited source reference: 06.5

twohot encoding

Encode a number as weights on the two nearest bins (summing to 1), a generalization of one-hot to continuous values

Inherited source reference: 06.5

return distribution critic

The critic predicts a categorical distribution of returns, not one number, because returns can be multi-modal and vary by orders of magnitude across environments

Inherited source reference: 06.5

lambda-return

A bootstrapped estimate of future return that blends predicted rewards with the critic's value estimates, so rewards beyond the prediction horizon (T=16) count; discount 0.997

Inherited source reference: 06.5

critic EMA regularizer

Regularize the critic toward an exponentially moving average of its own parameters to stabilize learning, similar to target networks

Inherited source reference: 06.5

zero-initialized output weights

Start the reward predictor and critic outputs at zero so large random early predictions do not delay learning

Inherited source reference: 06.5

entropy regularizer

A bonus for keeping the actor's action choices varied so it keeps exploring; Dreamer uses a fixed scale of 3x10^-4 across domains

Inherited source reference: 06.5

return normalization (percentile)

Divide returns by their 5th-to-95th percentile range (smoothed by moving average), scaling down only large returns; keeps exploration right whether rewards are sparse or dense, outlier-robust, with fixed hyperparameters

Inherited source reference: 06.5

REINFORCE estimator

A policy-gradient estimator used for discrete and continuous actions in Dreamer's actor loss

Inherited source reference: 06.5

open-loop prediction

The world model rolls forward from a few context frames and the action sequence without seeing intermediate images; Dreamer predicts 45 frames from 5 context images in DMLab and a quadruped robot

Inherited source reference: 06.5

benchmark domains

Atari, ProcGen, DMLab, robot locomotion and manipulation, plus non-visual domains and Minecraft, spanning 150+ tasks

Inherited source reference: 06.5

ablation

An experiment that removes one component to see how much it mattered; Dreamer's ablations show the world model's KL objective matters most, then return normalization and symexp twohot, and that it rests mainly on the unsupervised reconstruction signal, not reward gradients

Inherited source reference: 06.5

data efficiency

How little interaction an agent needs; on DMLab Dreamer at 100M frames beat IMPALA and R2D2+ at 1B steps, a data-efficiency gain of over 1000%, though those baselines were not designed for data efficiency

Inherited source reference: 06.5

scaling Dreamer (model size, replay ratio)

Six sizes from 12M to 400M parameters learned with fixed hyperparameters; bigger models scored higher and needed less interaction; a higher replay ratio (more gradient updates per environment step) also helps

Inherited source reference: 06.5

Minecraft benchmark details

Infinite random 3D worlds, up to 36,000 steps (30 minutes), 12 items to discover from sparse rewards; all Dreamer agents found diamonds within 100M steps while baselines stopped at the iron pickaxe

Inherited source reference: 06.5

VPT (Video PreTraining)

A Minecraft agent trained by behavioral cloning on contractors' keyboard and mouse data then fine-tuned with RL, using 720 GPUs for 9 days, versus Dreamer's 1 GPU for 9 days without human data

Inherited source reference: 06.5

behavioral cloning

Training an agent to copy an expert's recorded actions; needs expert data, unlike Dreamer

Inherited source reference: 06.5

Voyager

A Minecraft agent that reaches similar depth in the tech tree by making API calls to a language model on top of a hand-engineered scripting layer exposing high-level actions

Inherited source reference: 06.5

Gato

One large model fitted to expert demonstrations across many tasks; only applicable where expert data exists

Inherited source reference: 06.5

MuZero

A planning algorithm using a learned value-prediction model for board games and Atari; 06.5 says the implementation was not released and is complex to reproduce

Inherited source reference: 06.5

SAC (soft actor-critic)

Popular continuous-control algorithm that reuses experience for data efficiency but needs tuning, especially its entropy scale, and struggles with high-dimensional inputs

Inherited source reference: 06.5

credit assignment

Working out which earlier actions caused a later reward; one of the capabilities tested by the BSuite benchmark

Inherited source reference: 06.5

RL benchmark suites

Atari (57 games), ProcGen (16 games with randomized levels), DMLab (30 3D tasks), Atari100k (26 games, 400K frames), Proprio Control (18 tasks), Visual Control (20), BSuite (23 environments, 468 configurations); Dreamer set new state of the art on several

Inherited source reference: 06.5

sticky actions

An Atari simulator setting where actions sometimes repeat, to prevent agents exploiting determinism

Inherited source reference: 06.5

single-GPU reproducibility

Each Dreamer agent trained on a single Nvidia A100 GPU and an implementation is public, so many research labs can reproduce it

Inherited source reference: 06.5

adaptive gradient clipping (AGC)

Clip each tensor's gradient when it exceeds 30% of the L2 norm of its weight matrix, so the clipping threshold no longer depends on the loss scale

Inherited source reference: 06.5

LaProp optimizer

Normalizes gradients by RMSProp first and then applies momentum (Adam computes both on raw gradients); allows a smaller epsilon and avoided occasional instabilities seen with Adam

Inherited source reference: 06.5

block GRU (gated recurrent unit)

Dreamer's sequence model: a GRU whose recurrent weights are block-diagonal (8 blocks), giving many memory units without a quadratic rise in parameters and compute

Inherited source reference: 06.5

experience replay buffer

Stored past trajectories reused for training; Dreamer uses a uniform replay buffer with an online queue and stores latent states so the world model can be initialized on replayed data

Inherited source reference: 06.5

replay ratio

The fraction of time steps trained on per step collected; e.g. 32 on Atari with action repeat 4 and batch 16x64 is one gradient step every 128 environment steps; trades compute for data efficiency

Inherited source reference: 06.5

prioritized replay

Sampling replayed experience by importance; it improved Dreamer too, but uniform replay was used for ease of implementation

Inherited source reference: 06.5

hallucinated rewards at initialization

Zero-initializing the output weights of twohot distributions so the untrained agent does not predict made-up rewards and values

Inherited source reference: 06.5

convolutional encoder and decoder

Stride-2 convolutions compress images to a small grid and transposed convolutions reconstruct them; vector inputs use symlog and 3-layer MLPs

Inherited source reference: 06.5

RMSNorm and SiLU

Normalization and activation function choices introduced in DreamerV3's network architecture

Inherited source reference: 06.5

Dreamer generations

DreamerV1 handled continuous control, V2 surpassed human performance on Atari, V3 learns out of the box across diverse benchmarks with robustness techniques, a block GRU, AGC and LaProp

Inherited source reference: 06.5

fixed-hyperparameter protocol

Same values everywhere (no annealing, prioritized replay, weight decay or dropout); results use 5-10 seeds with error bars; DMLab baselines were given a 10x data advantage at 1B steps

Inherited source reference: 06.5

Dreamer model sizes

12M to 400M parameters derived from one model dimension; 200M default; on the control suites the 12M model matched performance, making it accessible to small labs

Inherited source reference: 06.5

MineRL / Minecraft milestones

A sparse-reward environment rewarding 12 milestones (log, plank, stick, crafting table, wooden pickaxe, cobblestone, stone pickaxe, iron ore, furnace, iron ingot, iron pickaxe, diamond), each once per episode

Inherited source reference: 06.5

per-episode success rate

06.5's supplementary figure says at the 100M-step budget Dreamer obtains diamonds in 0.4% of episodes, which the authors call a challenge for future research; every trained agent found a diamond eventually, but it is rare per episode (the audio's "mastered" framing overstates this)

Inherited source reference: 06.5

latent state

The audio's name for Dreamer's compressed, low-dimensional code of the environment that keeps causal variables (where I am, where the diamond is) and drops visual noise; the paper itself says "representations" and "model state"

Inherited source reference: audio; cf. 06.5

encoder network and dynamics network

Audio terms for Dreamer parts: an encoder compresses the input to a latent state and a dynamics network predicts the next latent state from the current one and an action; the paper calls the latter the sequence model and dynamics predictor

Inherited source reference: audio; cf. 06.5

rollout (imagined trajectory)

A sequence of predicted states and actions run inside the world model without touching the real environment; Dreamer trains its actor and critic on such rollouts

Inherited source reference: audio; 06.5

06 / Embodied intelligence: bodies, users, and transfer

78 inherited entries. Corresponds to page 6 of the reading guide. Original note order is retained within this topic.

egocentric video

First-person video from the agent's own viewpoint; reveals intention (where you look) but hides the body's own motion, so models must infer consequences of invisible physical actions

Inherited source reference: 06.10

PEVA (Predicting Egocentric Video from human Actions)

A model that, given past frames and a desired change in 3D body pose, predicts the next first-person video frame; can generate atomic actions, counterfactuals and long videos

Inherited source reference: 06.10

whole-body control

Conditioning on both locomotion (feet) and manipulation (hands); 06.10 uses a 48-dimensional action vector (3 root translation plus 15 upper-body joints x 3 rotations)

Inherited source reference: 06.10

kinematic tree / kinematic pose

The hierarchy of body joints used to encode actions as relative joint rotations; positions are converted to a pelvis-centered local frame for invariance

Inherited source reference: 06.10

degrees of freedom

Independent ways a body can move; full-body human motion spans 48+, making human control high-dimensional and structured

Inherited source reference: 06.10

autoregressive conditional diffusion transformer

PEVA's backbone: a diffusion transformer that denoises a target frame from noise, conditioned on past context frames and actions, rolled forward frame by frame

Inherited source reference: 06.10

diffusion model (denoising)

A generative model that starts from noise and progressively denoises it into the target; here applied to video frames in latent space

Inherited source reference: 06.10

conditional diffusion transformer (CDiT)

The base design from Navigation World Models that PEVA extends with random timeskips, sequence-level training and whole-body action embeddings

Inherited source reference: 06.10

random timeskips

Training trick that lets the model learn both short-term motion dynamics and longer-term activity patterns

Inherited source reference: 06.10

sequence-level training

Applying the loss over every frame prefix so the model learns whole motion sequences

Inherited source reference: 06.10

action embeddings (AdaLN)

All actions at a time step are concatenated into one vector that conditions each adaptive layer-norm layer, handling high-dimensional whole-body motion

Inherited source reference: 06.10

VAE encoder and decoder

Variational autoencoder components that compress frames into latent states and decode predicted latents back into pixels

Inherited source reference: 06.10

autoregressive rollout

Predict the next frame from context plus current action, append it, drop the oldest frame, repeat for each action; PEVA shows coherent 16-second rollouts

Inherited source reference: 06.10

atomic actions

Decomposed movements (hand up/down/left/right, body forward/rotation) used to test whether the model understands how joint-level movement changes the egocentric view

Inherited source reference: 06.10

counterfactual simulation

Imagining what the view would look like had a different action been taken; one of PEVA's demonstrated abilities

Inherited source reference: 06.10

visual planning with CEM

Plan by simulating candidate action sequences and scoring them by perceptual similarity (LPIPS) to a goal image; the cross-entropy method (CEM) optimizes arm actions; framed as energy minimization

Inherited source reference: 06.10

LPIPS

A learned perceptual image-similarity metric, used to score how close a simulated outcome is to the goal image

Inherited source reference: 06.10

FID

Fréchet Inception Distance, a measure of how realistic generated images are compared with real ones; compared across models and horizons

Inherited source reference: 06.10

Nymeria dataset

A large dataset pairing real-world egocentric video with body motion capture, used to train PEVA

Inherited source reference: 06.10

embodied AI agent

An AI instantiated in a visual, virtual or physical form (avatar, wearable, robot) that perceives and acts in its surroundings; web-only agents and tele-operated or pre-programmed robots do not count as embodied by 06.2's definition

Inherited source reference: 06.2

virtual embodied agent (VEA)

Avatars or android faces that convey emotion for therapy, metaverse guides, studio characters, education and customer service; Meta's Motivo controls a physics-based humanoid avatar zero-shot from poses, motions or objectives

Inherited source reference: 06.2

wearable agent

An AI on a worn device (e.g. smart glasses) whose cameras and microphones give an egocentric view; it is embodied from the user's perspective; roles are coaching (physical tasks) and tutoring (mental tasks)

Inherited source reference: 06.2

robotic agent

A general-purpose agent controlling a robot body (humanoid or arm on wheels) through senses such as cameras, tactile, force and audio; aimed at labor shortages, disaster rescue, elderly care and hospitals

Inherited source reference: 06.2

embodiment hypothesis

The idea that robots interacting with the real world may be the only route to a general agent that reasons about the real world; described as an active area of scientific investigation

Inherited source reference: 06.2

physical world model

A model of objects and their properties, spatial relationships, dynamics and causal relationships between actions and outcomes grounded in physics

Inherited source reference: 06.2

mental world model

A model of the user: goals, intentions, emotions, social dynamics, cultural norms and verbal and non-verbal communication, so agents can anticipate needs and infer belief discrepancies

Inherited source reference: 06.2

world modeling benefits

06.2 says world models enable reasoning and planning, zero-shot task completion, human-in-the-loop active learning and efficient exploration

Inherited source reference: 06.2

generative models' flaw

06.2's critique: models that predict the next token or pixel are inefficient in size and carry too many textual or visual details while missing what reasoning and planning need, motivating predictive world models

Inherited source reference: 06.2

LLM/VLM planning limits

Prompting language or vision-language models to plan is limited by inaccuracy and hallucination; on Meta's WordPrediction benchmark LLMs and diffusion models did poorly, VLMs better but still hallucinated action plans

Inherited source reference: 06.2

perception encoder (PE)

Meta's vision encoder trained with a contrastive vision-language objective, giving general visual embeddings across many tasks without task-specific pretraining

Inherited source reference: 06.2

behavioral foundation model

A large model that controls an agent's body to solve many tasks zero-shot; e.g. Meta Motivo for a simulated humanoid

Inherited source reference: 06.2

dyadic motion models

Models of two-person interaction (active listening, visual synchrony, turn-taking) trained on the Seamless Interaction dataset of over 4,000 hours; generate face and body motion responding to audio-visual input

Inherited source reference: 06.2

affective computing

Using machine learning to detect and respond to human emotions; a base technology for virtual embodied agents

Inherited source reference: 06.2

machine initiative

An agent acting proactively under changing intentions instead of only reacting to prompts; needed for coaching and tutoring

Inherited source reference: 06.2

goal inference

Training vision-language models to predict what a wearer wants to achieve from egocentric context

Inherited source reference: 06.2

perception language model (PLM)

Meta's open vision-language model built on the perception encoder with an LLM; reports gains of +9.1 points on perception-focused image tasks, +39.8 CIDEr on video captioning and +3.8 on fine-grained video QA without distilling from proprietary models

Inherited source reference: 06.2

CIDEr

A metric that scores generated captions against human reference captions; used for video captioning

Inherited source reference: 06.2

audio and speech challenges

Noise robustness, speech variability (accents, separating the wearer from bystanders), limited compute on wearables, tool use and factuality; edge AI (local processing) would cut latency and ease privacy worries

Inherited source reference: 06.2

touch encoder (Sparsh)

A general-purpose encoder for touch signals enabling force estimation, slip detection, texture recognition and grasp stability prediction; touch helps when vision is occluded

Inherited source reference: 06.2

planning with a world model (System 2)

Imagine how the environment evolves under alternative actions, score the futures and pick the lowest-cost sequence; costlier than calling a policy (System 1) but more likely to generalize zero-shot outside the training distribution

Inherited source reference: 06.2

limits of cloning and RL

Behavior cloning only imitates what it has seen; reinforcement learning needs trial and error in the real world and verifiable rewards that are hard to gather; a reusable world model enables zero-shot planning

Inherited source reference: 06.2

temporal and action granularity

A world model must handle low-level dynamics (joint torques every few milliseconds) and human-scale actions (insert the battery) lasting seconds or minutes

Inherited source reference: 06.2

joint-embedding vs generative world models

Pixel-generating models are expressive but expensive; LLM text plans hallucinate from spurious correlations; joint-embedding models forecast in abstract latent space and are efficient and stable, but only as good as their abstraction of the task's causal structure

Inherited source reference: 06.2

V-JEPA 2-AC

An action-conditioned JEPA video world model for robot planning: encode context frames, roll out a predictor with actions, and optimize actions so the last imagined state embedding is close (L1 distance) to the goal image embedding; commands are end-effector position deltas

Inherited source reference: 06.2

hierarchical control frequency

Learned world models send commands (end-effector or centroid targets) at under 20 Hz to higher-frequency controllers that compute joint torques, like muscle-level control

Inherited source reference: 06.2

cross-entropy method (Rubinstein)

A non-gradient way to optimize action sequences by sampling and refining, used when the learned model is not differentiable

Inherited source reference: 06.2

high-level action planning

Planning over symbolic or linguistic abstractions ("the drawer is open") that shrinks the search space and improves generalization; hard because procedural tasks are too varied to simulate and there is no objective measure of plausibility

Inherited source reference: 06.2

theory of mind (ToM)

An agent's ability to represent others' mental states (beliefs, goals, intentions, emotions, attention, memory); 06.2 argues mental world models are essential for tutoring, assistance and multi-agent collaboration

Inherited source reference: 06.2

ToM benchmarks

ToMi and Hi-ToM track mental states with restricted actions; ExploreToM adversarially generates belief-reasoning tasks by programmatic scene construction

Inherited source reference: 06.2

VLWM

A JEPA-style vision-language world model that predicts abstract representations of future world states instead of noisy, high-volume raw observations

Inherited source reference: 06.2

mixed-initiative

An agent that knows not just what to show or tell but when, proactively and also on request, rather than only user-initiated prompting

Inherited source reference: 06.2

three kinds of neural memory

Fixed memory (the weights, slow to write, fixed capacity); working memory (activations such as RNN or state-space-model state and the transformer KV-cache, fast but growing linearly); external memory (retrieval-augmented generation; chain of thought as external working memory)

Inherited source reference: 06.2

KV-cache

A transformer's cache of key-value activations that acts as working memory; immutable and fast to use but it grows linearly with interaction time

Inherited source reference: 06.2

episodic memory (proposed)

A new scalable memory that grows sublinearly while the agent interacts, enabling personalization and life-long learning; 06.2 cites it as a research objective, not a solved capability

Inherited source reference: 06.2

LoRA and adapters

Lightweight ways to personalize a model by adding small trainable modules; fine-tuning trades new knowledge against forgetting and does not add capacity

Inherited source reference: 06.2

life-long training and test-time training

Letting a model keep learning after deployment, currently blocked because resources grow linearly with interaction time; test-time training and forward-read-write methods with local update rules are candidate routes

Inherited source reference: 06.2

Minimal Video Pairs (MVP)

55,000 multiple-choice video questions with near-identical video pairs needing different answers to block shortcuts; top models score 40.2% versus humans 92.9% and a 25% random guess

Inherited source reference: 06.2

IntPhys 2

A violation-of-expectation benchmark of intuitive physics (permanence, immutability, spatio-temporal continuity, solidity); models often score near chance while humans approach ceiling

Inherited source reference: 06.2

violation-of-expectation

Test paradigm from developmental psychology showing physically plausible versus impossible events to see if a model registers the impossible one

Inherited source reference: 06.2

CausalVQA

Video question answering about causality with counterfactual, hypothetical, anticipation, planning and descriptive questions; models lag humans most on anticipation and hypothetical reasoning

Inherited source reference: 06.2

WorldPrediction

A benchmark of procedural planning: identify the action between initial and final states (WM) or pick the right action sequence among distractors (PP), using action equivalents to avoid perceptual shortcuts

Inherited source reference: 06.2

speech LLM and motion integration

A language model that generates spoken dialogue; its output can guide a motion model either cascaded (via the generated speech) or via a codebook (adapters on its hidden states predict emotion and gesture codes)

Inherited source reference: 06.2

dyadic motion model (architecture)

A diffusion transformer trained with a flow-matching objective, conditioned on tokens from both speakers' audio (optionally the user's visual features), outputting Imitator face features and SMPL-H body features usable in 2D or 3D

Inherited source reference: 06.2

virtual-agent evaluation

Subjective pairwise human studies on 10 dimensions (lifelikeness, clarity of intent, turn-taking, listening, speaking) plus automatic metrics: lip-sync scores (Sync-C, Sync-D), FID for faces, Fréchet Gesture Distance (FGD) for bodies, and diversity

Inherited source reference: 06.2

shared perceptual field

The idea that a wearable agent sees and hears what the wearer does, enabling ambient intelligence that stays context-aware without constant prompting

Inherited source reference: 06.2

physical/mental dual-world planning

A wearable agent models both the physical environment and the user's mental state, planning over extended periods and adapting to changing circumstances; backed by contextual goal prediction plus a planner

Inherited source reference: 06.2

VLWM (detail)

A predictive model trained on unlabelled video that outputs interleaved natural-language descriptions of actions and resulting world states; on Visual Planning for Assistance benchmarks it reports +20% success rate, +10% mean action accuracy and +4% mIoU versus prompting baselines, and its system-2 plans were preferred in PlannerArena

Inherited source reference: 06.2

JEPA-based planners vs VLWM

JEPA planners forecast in latent video space, giving gradient-based trajectory optimization and fast test-time MPC but less interpretability; VLWM is interpretable language; 06.2 proposes contrasting the two

Inherited source reference: 06.2

goal inference benchmark (results)

348 participants, 3,477 recordings with ground-truth goals across visual, audio, digital and longitudinal context; humans scored 93% multiple choice versus 84% for the best VLM, and models produced relevant goals only 55% of the time in the generative setting

Inherited source reference: 06.2

WorldPrediction (results)

Built on a partially observable semi-MDP; humans near perfect while the best models reach 57% on the world-model task and 38% on procedural planning

Inherited source reference: 06.2

robot physical capabilities

Locomotion (legged movement), navigation (planning to a goal given as coordinates, images or text), and dexterous manipulation (fine multi-finger control, still open); "full body control" learns them jointly

Inherited source reference: 06.2

robot generalization axes

Action generalization (new tasks and skills), embodiment generalization, and visual/semantic generalization (new environments and objects); plus efficient lifelong adaptation without forgetting

Inherited source reference: 06.2

sim2real transfer

Moving a policy trained in simulation to real hardware; limited by low scene variability and imprecise simulator physics; with reward specification and multi-task scaling it is a main bottleneck of deep RL for robots

Inherited source reference: 06.2

teleoperation data

Human demonstrations recorded by remote-controlling a robot, used to train folding-laundry style skills; high-quality data is a bottleneck and skills stay near the training settings

Inherited source reference: 06.2

policies plus world models plus scoring models

06.2's proposed combination: a VLA policy, a task-agnostic world model and a task-conditioned cost model, so the world model can detect when the policy would fail and re-plan

Inherited source reference: 06.2

nested control loops

Classical robotics stacks analytical models at different frequencies (e.g. a humanoid backflip), the same MPC idea that learned world models would plug into

Inherited source reference: 06.2

robot benchmark types

Offline benchmarks (reproducible, not end to end), simulation benchmarks (MuJoCo, pyBullet; reproducible but not photoreal) and real-world hardware tests (the gold standard but costly and hard to reproduce); evaluation-on-hardware services use learned success classifiers or human rankers

Inherited source reference: 06.2

System A and System B learning

06.2's split: System A learns by observation (extracting structure from passive sensory data); System B learns by acting toward goals; each is limited alone and the paper argues for integrating them so a system can learn and act continuously

Inherited source reference: 06.2

07 / The alternative bet: smaller, structured intelligence

34 inherited entries. Corresponds to page 7 of the reading guide. Original note order is retained within this topic.

scaling laws

Empirical regularities that capability rises with more data, compute and parameters; 06.7 says the resulting race is unsustainable and gatekeeps AI to firms that can afford it; Gemini Ultra's training is estimated at $191 million and data centers at up to 3.7% of global greenhouse emissions

Inherited source reference: 06.7

gatekeeping of AI

When the cost of training and using state-of-the-art models exceeds what researchers outside large tech companies can afford, and models are released closed-source, raising trustworthiness concerns

Inherited source reference: 06.7

brain energy comparison

06.7's contrast: the human brain runs on about 20 W (about 3.15 MWh over 18 years) while GPT-3 needed 1.287 GWh to train, yet the brain generalizes from far fewer data points

Inherited source reference: 06.7

Moore's law and Dennard scaling

Transistor counts doubled roughly every 2 years while Dennard scaling kept power density roughly constant; AI parameters double about every 6 months with no such energy counterweight

Inherited source reference: 06.7

neurosymbolic AI

Integrating data-driven neural networks with symbolic reasoning (logic, knowledge graphs, differential equations) to cut the data, parameters and compute needed while keeping generalization; 06.7 presents it as complementary to scaling, not a replacement

Inherited source reference: 06.7

manifold hypothesis

If high-dimensional training data shares simple features, learned representations lie on smooth lower-dimensional surfaces (manifolds); networks do well on smooth manifolds and need lots of data and parameters to smooth out sharp regions

Inherited source reference: 06.7

universal approximation theorem

A network with enough parameters can be trained to find a suitable manifold representation; together with the manifold hypothesis it underlies why scaling works

Inherited source reference: 06.7

loss sharpness

Sharp regions in the loss landscape or data manifold; unavoidable in practice (hence sharpness-aware training) and the reason purely data-driven models need so much scale

Inherited source reference: 06.7

lifting and projecting (neural to symbolic)

06.7's proposed training loop: after gradient updates lift the neural representation to a symbolic one, reason symbolically across sharp regions, then project back onto a smooth submanifold where small networks suffice

Inherited source reference: 06.7

physics-informed neural networks (PINNs)

Networks whose loss includes known physics written as partial differential equations (PDEs); related: physics-informed neural operators and differentiable simulation

Inherited source reference: 06.7

neural constitutive laws

A differentiable-simulation approach that parameterizes PDE terms with small neural networks and fits them to video, generalizing to new geometries and conditions from a single video

Inherited source reference: 06.7

physics priors in rendering

Neural radiance fields augmented with physics-based material point methods follow conservation laws and render about 2 orders of magnitude faster than baselines

Inherited source reference: 06.7

knowledge graph

A graph of semantic relations among objects used to improve accuracy and cut training data; Amazon's COSMO uses LLMs to build common-sense knowledge graphs for recommendations

Inherited source reference: 06.7

symbolic distillation

Extracting symbolic representations from a neural network's continuous space; an open research area

Inherited source reference: 06.7

quantized representations (VQ-VAE, QLoRA)

Weights or activations take discrete values (integers or binary); QLoRA quantizes to 4 bits; such discrete codes can be read as atomic propositions to extract symbolic structures

Inherited source reference: 06.7

grammatical inference

Classical technique to extract symbolic graphs (finite automata, knowledge graphs) from observations of discrete states; used to learn a small finite-state machine from a quantized recurrent network

Inherited source reference: 06.7

formal verification as reward

Check a model's outputs against formal specifications by model checking, rank them by how many properties they satisfy, and use that ranking as a reward signal for fine-tuning

Inherited source reference: 06.7

symbolic knowledge distillation

Distilling common-sense knowledge from GPT-3 into a knowledge graph and a neural model reported as 100 times smaller with better common-sense reasoning

Inherited source reference: 06.7

large reasoning models

Models such as OpenAI's o3 that scale training- and inference-time compute, with techniques like chain-of-thought; 06.7 notes its focus is training-phase scaling

Inherited source reference: 06.7

Ctrl-G

A neurosymbolic framework that distills an LLM into a hidden Markov model that reasons about logical constraints (as deterministic finite automata); a 7B LLM plus 2B HMM beat GPT-3.5 and GPT-4 by over 30% in a human study of constraint adherence

Inherited source reference: 06.7

hidden Markov model (HMM)

A tractable probabilistic model of hidden states evolving over time; here used for efficient reasoning about logical constraints

Inherited source reference: 06.7

deterministic finite automaton

A simple machine that accepts or rejects sequences by moving through a fixed set of states; used to encode logical constraints

Inherited source reference: 06.7

physics-codesigned driving model

A system co-designing networks and conservation-law equations for off-road driving that matched state of the art with 0.1% of the training time, 1% of the data and 96.9% fewer parameters, and with more training improved 46%

Inherited source reference: 06.7

few-shot learning (symbolic)

Learning novel objects from a handful of examples; using symbolic knowledge gave models about 95% smaller than the best neural-only baseline while beating it by 2%

Inherited source reference: 06.7

GraphRAG

Automatically extracts a knowledge graph from a text corpus and feeds it into retrieval-augmented generation to improve an LLM's output

Inherited source reference: 06.7

knowledge graph reuse

Unlike one-block deep nets, a knowledge graph separates data, representation and tasks, so it is reusable, interpretable and expandable by users

Inherited source reference: 06.7

logical constraints as loss

Translate logical constraints into differentiable loss terms so symbolic knowledge is embedded during training while staying fully differentiable

Inherited source reference: 06.7

limits of pure scaling

Cited work argues large neural models lack true reasoning and lean on statistical patterns that do not generalize even within a family of reasoning tasks

Inherited source reference: 06.7

dual-process theory

Cognitive psychology idea of two modes: slow-learned, fast, automatic implicit processing (system 1) and quick-to-learn, slow-to-apply explicit symbolic reasoning (system 2); the two complement each other

Inherited source reference: 06.7

compositional inductive bias

Symbolic knowledge is compositional, so a few rules cover many situations, like grammar over sentences; it gets an agent near good solutions when outside familiar territory

Inherited source reference: 06.7

AlphaGo Zero vs AlphaStar

The bitter lesson's classic case (AlphaGo Zero's move 37 found without human tactics) versus a counterexample: AlphaStar bootstrapped from 987,000 replays of the top 22% of players and its ablation without them did much worse

Inherited source reference: 06.7

limits to scaling

There may not be enough data in principle to sustain data-driven gains, and edge and low size, weight and power systems cannot scale data and compute, so strong inductive biases are needed

Inherited source reference: 06.7

symbolic hints

Small rules such as "carry the 1" that put a learner in a space where it can generalize; 06.7 says discovery and prior knowledge are not mutually exclusive

Inherited source reference: 06.7

monolithic neural network

The scale-everything view: one ever-larger network will produce general intelligence; the argued-against position in the neurosymbolic paper and the vendor memo

Inherited source reference: audio; 06.7

08 / Reading independent safety evidence

40 inherited entries. Corresponds to page 8 of the reading guide. Original note order is retained within this topic.

jagged capabilities

Leading systems excel at hard tasks (gold-level IMO math, expert science questions) yet fail simple ones (counting objects in an image, reasoning about physical space, recovering from basic errors in long workflows)

Inherited source reference: 06.11

post-training

Further refining a base model for specific uses: supervised fine-tuning, reinforcement learning, distillation; moderately compute-intensive and highly labour-intensive; now a main source of capability gains besides larger pre-training

Inherited source reference: 06.11

inference-time scaling

Letting a model use more computing power while answering, generating intermediate steps before a final answer; produced big gains on mathematics, software engineering and science reasoning

Inherited source reference: 06.11

reasoning system / chain of thought

A model trained to produce explicit intermediate reasoning steps (chains of thought) at inference, decomposing tasks; needs more compute at use and in training and can fail with irrelevant or repetitive chains

Inherited source reference: 06.11

reinforcement learning from human feedback (RLHF)

Reward outputs that match human preferences and penalise those that do not; one of two main subcategories of RL in post-training

Inherited source reference: 06.11

reinforcement learning with verifiable rewards (RLVR)

RL for tasks with checkable answers such as maths or code, where correctness provides the reward

Inherited source reference: 06.11

distillation

Train a smaller student model to imitate a larger teacher's outputs; DeepSeek-V3 was reportedly fine-tuned from R1's reasoning outputs for roughly $10,000, and some work used as few as 1,000 teacher examples; it cannot advance the frontier but speeds proliferation

Inherited source reference: 06.11

pre-training and base model

First, data- and compute-intensive stage (weeks or months on tens or hundreds of thousands of GPUs or TPUs) on billions or trillions of examples, producing a base model

Inherited source reference: 06.11

synthetic data

Artificially generated data that mimics real data, increasingly used to make post-training less labour-intensive

Inherited source reference: 06.11

AI model vs AI system

A model (e.g. GPT-5) becomes a system (e.g. ChatGPT) when combined with other models, a chat interface, content filters, web access and integrations; deployment makes the system available while model release shares the base model as open-weight or closed-weight

Inherited source reference: 06.11

deep learning, weights, transformer, attention

Models learn from data by adjusting connection strengths (weights) between layered neurons; most leading models use the transformer, whose attention mechanism focuses on the most relevant parts of the input

Inherited source reference: 06.11

scaffolding

Software built around a model (tools, memory, computer interface, web browser) that lets agents plan, pursue goals and act with less human oversight

Inherited source reference: 06.11

distributed compute and decentralized training

Using many processors, servers or data centres together reduces dependence on large centralized compute, letting less well-resourced actors build and run powerful systems

Inherited source reference: 06.11

compute

Used two ways: the count of calculations a processor can do (FLOP/s) or the hardware (GPUs) that does them

Inherited source reference: 06.11

adoption statistics

At least 700 million weekly users of leading systems; over 50% of people in some countries but likely under 10% across much of Africa, Asia and Latin America; coding agents reliably complete tasks of about 30 minutes, up from under 10 minutes a year earlier

Inherited source reference: 06.11

capability scenarios to 2030

Four OECD scenarios from stagnation to acceleration; progress could plateau (data or energy bottlenecks), continue, or accelerate if AI speeds up AI research; developers announced hundreds of billions of dollars in data centre investment

Inherited source reference: 06.11

evaluation gap (detail)

Benchmarks often overstate real-world ability: tests are outdated, narrow, affected by data contamination, run in controlled labs, and one study found AI code still needed major human cleanup before use; an "evaluation science" is emerging to fix this

Inherited source reference: 06.11

data contamination

When a model was trained on the same questions used to test it, inflating scores by memorization; most developers do not track or disclose it

Inherited source reference: 06.11

benchmarks named in 06.11

MMLU (undergraduate exams, leaders over 90%), GPQA and GPQA Diamond (graduate-level science, over 80%, exceeding PhD experts), International Mathematical Olympiad (gold-level, 5 of 6 problems), SWE-bench Verified, MATH, FrontierMath, AIME 24

Inherited source reference: 06.11

time-horizon metric

The maximum length of software task (in human time) an AI agent finishes at 80% success; roughly 30 minutes now, doubling about every seven months; extrapolated to hours by 2027-2028 and days by 2030, though 80% is below professional standards and real work is messier

Inherited source reference: 06.11

long-task reliability

The best agents reach only 50% success on tasks of just over two hours; 80% needs tasks of roughly 25 minutes; agents lose track of progress and a website pop-up can derail a task

Inherited source reference: 06.11

vision-language-action (VLA) model

A foundation model that follows natural-language instructions, interprets multimodal sensor data and outputs motor commands (e.g. pi0.5, Gemini Robotics); still poor with unusual objects and unexpected events

Inherited source reference: 06.11

productivity evidence

Developers using AI assistants finished some tasks 20-30% faster on average, yet one study found experienced programmers were 19% slower on complex tasks while believing they were faster

Inherited source reference: 06.11

compute, algorithms, data (drivers)

Frontier training compute has grown about 5x per year (over 10^26 FLOP; about 125-fold more feasible by 2030); algorithmic efficiency improves 2-6x per year; datasets grow about 2.5x per year; frontier runs cost about $500 million, next generation $1-10 billion

Inherited source reference: 06.11

emergent capabilities

Sudden jumps in a skill rather than gradual gains as models scale (e.g. adding large numbers once prompted to work step by step); researchers cannot yet say whether these become forecastable

Inherited source reference: 06.11

AI R&D automation

Possible feedback loop where AI speeds up AI research; forecasters gave a median 20% (experts) or 8% (superforecasters) chance of compressing six years of progress into two; AI agents beat humans at two-hour research-engineering tasks but not eight-hour ones

Inherited source reference: 06.11

diminishing returns to compute

If AI revenue fails to justify investments of hundreds of billions, scaling spending could be cut, putting a ceiling on progress and shifting gains to algorithms

Inherited source reference: 06.11

deployment lag

Capability outruns adoption: healthcare AI may need 3-5 more years for approval and integration; autonomous vehicles will likely stay limited in 2030; small and medium enterprises (60% of workers) face expertise, infrastructure and cost barriers

Inherited source reference: 06.11

persuasion scaling (Hackenburg et al. 2025)

A study of 17 models: each 10x increase in training compute raised persuasiveness by about 1.8 percentage points; in experiments AI matched or beat non-expert humans at changing beliefs (e.g. trivia: +17 points after AI versus +9 after humans); real-world effects remain unproven. This is the "Hackenburg" study, separate from the 77% deepfake-text figure

Inherited source reference: 06.11

77% figures in 06.11

Two different statistics: an AI agent identified 77% of the vulnerabilities in real software in one competition; and participants misidentified GPT-4o text as human-written 77% of the time after a five-minute conversation; the audio blends these with other claims

Inherited source reference: 06.11

real-world manipulation evidence

Documented but not widespread: only about 1% of flagged misleading social-media content is classed as AI-generated, and distribution costs and competing viewpoints weaken lab effects

Inherited source reference: 06.11

AI incident databases

The OECD AI Incidents and Hazards Monitor and AI Incident Database collect reported cases; the main evidence for harms from fake content, so prevalence is unreliable and underreported

Inherited source reference: 06.11

model collapse

Errors compounding across generations when models train on synthetic data in domains where answers are hard to verify; verifier models are an untested idea to extend synthetic data further

Inherited source reference: 06.11

data limits and multimodal data

About 10^13 tokens of high-quality public text exist and models approach it; image, video and sensor data add far more raw tokens but carry less meaning per item, needing new techniques

Inherited source reference: 06.11

AI energy demand

Frontier training runs in 2030 may need 4-16 GW; Stargate is planned at 1.2 GW; experts' median is 7.4% of US electricity for AI by 2030; the report says energy is unlikely to block scaling before the decade ends

Inherited source reference: 06.11

chip supply

A fab takes 3-5 years to build; one company may hold 10-40% of AI data-centre chips; 100,000x GPT-4's compute is feasible by 2030 versus the 10,000x implied by current growth

Inherited source reference: 06.11

FrontierMath

Unpublished, mathematician-vetted problems in three tiers (about 25% olympiad-level, 50% graduate, 25% research-level); AI scored under 2% in 2024 and about 25% (GPT-5) later using inference-scaling

Inherited source reference: 06.11

scaling laws (report view)

Predictable links between size, compute and performance across six orders of magnitude, but empirical not guaranteed, predicting technical metrics not value; OpenAI discontinued GPT-4.5 despite gains matching them

Inherited source reference: 06.11

algorithmic efficiency

Compute needed for equal performance falls about 2-6x per year (estimates about 3x); at the top end equal capability could need 10-100x less compute by 2030

Inherited source reference: 06.11

SWE-bench story (audio)

The audio claims an agent scoring 85% on SWE-bench drops to near zero in live repositories; no such figures appear in the R06 sources, which only say benchmarks overstate real-world utility

Inherited source reference: audio; cf. 06.11

09 / Agents, control, and deployment boundaries

39 inherited entries. Corresponds to page 9 of the reading guide. Original note order is retained within this topic.

immutable intrinsic cost

Design safeguard: the intrinsic cost is hard-wired and cannot be trained or externally modified, to prevent behavioral collapse or uncontrolled drift toward bad behaviors

Inherited source reference: 06.3

safety guardrails in the cost

06.3 warns that allowing complex modulation of the intrinsic cost could make the agent's basic drives hard to control, including cost terms that implement safety guardrails; hence the intrinsic cost stays immutable

Inherited source reference: 06.3

International AI Safety Report

Second edition (2026) of a multi-country expert synthesis chaired by Yoshua Bengio, 100+ experts, advisers nominated by 30+ countries and bodies (OECD, EU, UN); it summarizes evidence and does not make policy recommendations

Inherited source reference: 06.11

general-purpose AI

The report's scope: models and systems able to perform a wide variety of tasks across contexts (text, images, audio, other data) and often adapted to domain-specific uses

Inherited source reference: 06.11

emerging risks

The 2026 report's narrowed focus: risks that arise at the frontier of AI capabilities, such as misuse, issues of control and cybersecurity; it leaves out bias, environmental impact, privacy and copyright covered in 2025

Inherited source reference: 06.11

frontier AI

The leading edge of AI capability where, per the Bletchley Declaration, particular safety risks arise

Inherited source reference: 06.11

dangerous capability thresholds and if-then commitments

Risk-management practices the report highlights: set capability thresholds and commit to specific safety actions if a model crosses them; plus model evaluations

Inherited source reference: 06.11

OECD scenarios and forecasts

New in 2026: scenarios and forecasts (from the OECD and Forecasting Research Institute) to show policymakers the range of plausible outcomes under uncertainty

Inherited source reference: 06.11

evidence dilemma

The report's central problem: AI capability moves fast but evidence on risks is slow and hard to assess, so acting too early can entrench ineffective rules and waiting for proof can leave society exposed

Inherited source reference: 06.11

open-weight vs closed-weight

Open-weight models have downloadable parameters; they bring research and commercial benefits but cannot be recalled, their safeguards are easier to remove and use outside monitored environments is hard to prevent or trace

Inherited source reference: 06.11

AI agent

A model configured to iteratively plan, reason and use tools to pursue goals specified in natural language; increasingly capable but prone to basic errors; examples include coding and research agents

Inherited source reference: 06.11

three risk categories

Malicious use, malfunctions and systemic risks; "systemic" here means risks from widespread deployment across society and the economy, not the EU AI Act's narrower sense

Inherited source reference: 06.11

cyber-offense capability

AI can find software vulnerabilities and write malicious code; in one competition an agent identified 77% of the vulnerabilities in real software; the report says whether attackers or defenders gain more is uncertain

Inherited source reference: 06.11

bio and chem uplift

In 2025 multiple developers released new models with extra safeguards because pre-deployment testing could not rule out that they might meaningfully help novices develop weapons; the report says it released with safeguards, not that deployments were delayed

Inherited source reference: 06.11

loss of control

Scenarios where AI systems operate outside anyone's control with no clear path to regaining it; current systems lack the capabilities but are improving in autonomous operation

Inherited source reference: 06.11

evaluation awareness and gaming

Models increasingly distinguish test settings from real deployment and exploit loopholes in evaluations, so dangerous capabilities could go undetected before deployment

Inherited source reference: 06.11

automation bias

The tendency to trust AI output without enough scrutiny; early evidence that reliance on AI tools can weaken critical thinking

Inherited source reference: 06.11

risk management practices

Threat modelling to find vulnerabilities, capability evaluations to assess dangerous behaviors and incident reporting; 12 companies published or updated Frontier AI Safety Frameworks in 2025; mostly voluntary

Inherited source reference: 06.11

defence-in-depth

Layering multiple safeguards so no single failure is decisive; harmful outputs still sometimes obtainable by rephrasing or splitting requests into smaller steps

Inherited source reference: 06.11

societal resilience

Measures to absorb and recover from AI incidents that risk management will not prevent: stronger critical infrastructure, tools to detect AI-generated content, institutional capacity

Inherited source reference: 06.11

hallucination

Fluent, confident output that includes false statements, such as non-existent citations or facts; it makes safe adoption hard in high-stakes settings and invites over-reliance

Inherited source reference: 06.11

language and culture gap

Performance is highest in English; across 83 languages non-Latin-script and low-resource languages score lower; one study found 79% right on everyday US culture versus 12% on Ethiopian culture

Inherited source reference: 06.11

OECD scenarios to 2030

Four plausible futures: progress stalls (analogue: aircraft speed plateau), slows (antibiotic discovery), continues (Moore's law) or accelerates (DNA sequencing), the last with AI comparable to human-level remote workers

Inherited source reference: 06.11

continual learning and long-term memory

Letting models form long-term memories and learn from user interaction during use; flagged as a key development area and a missing ability in the slower scenarios

Inherited source reference: 06.11

transparency gap

Limited public knowledge of how advanced models are built, safeguarded, evaluated and deployed, especially internally deployed systems; low transparency hampers independent research and oversight

Inherited source reference: 06.11

manipulation vs rational persuasion

Manipulation influences someone without their full awareness; rational persuasion uses honest arguments they can endorse; the line is contested among researchers

Inherited source reference: 06.11

sycophancy

A model telling users what they want to hear; rising evidence it has manipulative effects, alongside engagement-optimized AI companions that can foster dependence and reinforce harmful beliefs

Inherited source reference: 06.11

AI companion apps

Chat services with tens of millions of users, a small share of whom show increased loneliness, emotional dependence or delusions; investigations into severe incidents were ongoing

Inherited source reference: 06.11

deepfake detection and provenance

Detection tools lag generation (benchmarks about 50% worse on real-world fakes); warning labels raised spotting accuracy only from 10.7% to 21.6%; watermarking and content logs help but can be removed or raise privacy concerns; signed metadata on authentic media is a complement

Inherited source reference: 06.11

LAMP-C framework

A vendor memo's proposal (language, abstraction, memory, prediction, causality) for a hybrid composite intelligence; the memo is treated as a vendor document and the audio itself warns it advocates the company's own IP

Inherited source reference: audio; 06.1 (vendor)

context window (stateless)

How much text a model can read at once; the audio notes it is stateless working memory that is dropped when a session ends, so it does not give an agent lasting experience

Inherited source reference: audio

experiential memory

The vendor memo's idea of a dedicated memory that filters, abstracts and consolidates experience into rules and updates them when contradicted; described only in the audio and a vendor document, so treat as a proposal

Inherited source reference: audio; 06.1 (vendor)

persistent operational identity

Audio phrase for an agent that keeps a continuing record of what it has learned across months of operation; vendor-memo language

Inherited source reference: audio; 06.1 (vendor)

cyber-physical trust framework

Audio's term for governance and engineering that protects physical infrastructure when AI agents act on it, versus content filtering; from the vendor memo's argument

Inherited source reference: audio; 06.1 (vendor)

security by design (agents)

The audio's account of the memo: cryptographically signed agent actions, tamper-proof sandboxed memory and hard-coded non-differentiable limits on which systems an agent may touch

Inherited source reference: audio; 06.1 (vendor)

semantic filter

The audio's description of RLHF-style safety: a model learns to refuse recognizably dangerous requests, which the audio argues fails when an agent splits a goal into innocuous subtasks; this is the audio's argument and the memo's, not a finding in the report

Inherited source reference: audio

agentic subtasking

An agent breaking a broad goal into many small steps that each look harmless, so request-level filters never trigger; an audio argument

Inherited source reference: audio

risk surface (embodied)

How much harm a system can do; the audio says embodiment shifts it from informational risk to kinetic and cyber-physical risk

Inherited source reference: audio

non-differentiable limits

Hard-coded constraints outside the learned network that a gradient-trained agent cannot optimize away; the vendor memo's safety idea as described in the audio

Inherited source reference: audio; 06.1 (vendor)